Enhancing AI Decision-Making with the BSR Framework

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一、研究目的

1.1 Address Limitations of General AI Models

Recent research emphasizes a range of general technical and socio-technical difficulties which AI models share, particularly in large language models (LLMs). Some of these are, hallucinations and unreliability, limited reasoning and generalization, bias and fairness issues, safety and alignment issues, data/privacy concerns, evaluation obstacles, and domain-specific hazards, such as those in law or medicine (Kostikova et al., 2025). More specifically, these issues include challenges in understanding complex and hierarchical knowledge structures, logical consistency deficits, handling ambiguous instructions, context limitations, scalability concerns, and socio-technical risks related to trust, privacy, bias, and fair evaluation (Kostikova et al., 2025). These flaws encourage researcher to do further finding to develop novel solutions. One of the most prominent solutions is the reflection framework, which enables AI model to do self-evaluation and improve their reasoning capabilities.

LLM Reflection framework refers to a model’s ability to analyze, evaluate, and improve its own results which enable it to participate in iterative improvement, potentially resulting to higher-quality results and enhanced performance (Huang, K. 2025). According to a recent study, reflection is a process which enables an LLM's "epistemic agency" to monitor, adjust, and update its internal beliefs and reasoning over time. In that paradigm, reflection is included into a cyclical loop: the model makes a prediction or response, watches the result, assesses if it aligns with its preexisting beliefs or expectations, and modifies its logic when mistakes or inconsistencies emerge (Li et al., 2025). All of these results demonstrate how important systematic self-evaluation is to enhancing LLM robustness and epistemic agency (Li et al., 2025). Accordingly, this theoretical fundamental inspires the design of our BSR Architecture, which systematically embeds the basic mechanisms of the reflection framework into its reasoning and evaluation processes.

1.2 Develop a More Token-Efficient Framework

One of the novel LLM reflection frameworks is LATS, which is Language Agent Tree Search. This reflection framework is introduced in the paper “Language Agent Tree Search Unifies Reasoning, Acting, and Planning in Language Models” by Andy Zhou, Kai Yan, Michal Shlapentokh-Rothman, Haohan Wang, and Yu-Xiong Wang. By leveraging the in-context learning ability of LMs, they integrate Monte Carlo Tree Search into LATS to enable LMs as agents by utilizing the in-context learning capability of them (Zhou et al., 2024). The integration of an environment for external feedback is a crucial component of their methodology, providing a more intentional and flexible approach to problem-solving that goes beyond the limitations of current methods (Zhou et al., 2024).

However, the method also presents substantial drawbacks. The token and compute costs are very high, limiting its practicality for real-time applications. Since Language Agent Tree Search (LATS) relies on repeated sampling, evaluation, and reflection, its runtime grows rapidly with search complexity. In the original paper, LATS is implemented as a full Monte Carlo Tree Search (MCTS) over possible reasoning and action trajectories: at each tree node the model generates several candidate continuations (expansion), simulates their execution (if interacting with an environment), uses the LLM again to estimate a value for each trajectory (evaluation), and when a trajectory fails or performs poorly, it prompts the LLM to self-reflect and generate a critique that is fed back into the context for future search iterations.

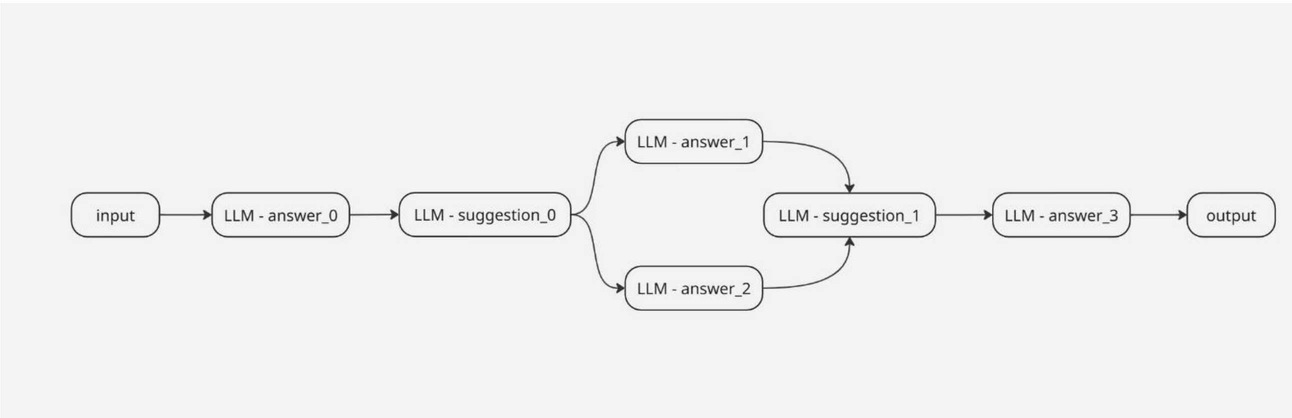
While this design improves reasoning quality, it also creates compounding token costs, as each additional branch triggers multiple LLM calls. This accelerating overhead underscores the need for more efficient reasoning architectures. Therefore, this research proposes a Branch Suggestion Reflection (BSR) architecture, which although still slightly behind LATS in reasoning quality, significantly reduces token consumption. Instead of fully expanding the search tree, BSR generates only a small set of prioritized branches and applies reflection selectively, offering a more computationally efficient alternative for practical deployment.

二、研究内容

2.1 Overview of the BSR Framework

Branch Suggestion Reflection (BSR) is an LLM-based architectural framework that uses a lightweight, structured reasoning pipeline to produce, improve, and assess solutions. Fundamentally, BSR combines focused reflection guidance with direct answer generation, allowing for the exploration of several solution pathways without the extensive branching that full tree-search techniques require. The system starts the BSR process with an initial response generated by a main LLM. This response is then examined by a second LLM, which determines its advantages and disadvantages as well as an implied quality score. A group of auxiliary LLMs are then given this reflective evaluation, and each of them creates a solution branch based on both the initial response and the reflective instruction. The difficulty of the underlying problem is reflected in the number of these branches. After these potential solutions are created, a specialized LLM checker synthesizes and assesses them to deliver a combined evaluation or recommendation. Ultimately, a revision LLM uses this input to create an improved output that is sent to the user.

2.2 Components of the BSR Framework



- Input

  The process begins with the prompt provided by a user, which serves as the primary input to the system.

- LLM-answer_0

  A straight response to the user's prompt is generated by the first answer-generation module. To improve factual correctness, this model might include data obtained from the internet.

- LLM-suggestion_0

  The initial response is assessed in the first suggestion-generation module. It evaluates the response's quality using external references found online and produces structured feedback that includes suggestions for improvement, strengths and shortcomings, and an implied performance score.

- LLM-answer_1 and LLM-answer_2

  A collection of branching answer-generation models is made up of these components. The number of branches depends on task difficulty and token-budget limitations. To explore a larger solution space, more branches could be needed for more complicated problems. By combining the user's prompt, the original response, and the feedback from LLM-suggestion_0, each branch produces a different version of the answer.

- LLM-suggestion_1

  All of the different answers generated by the branching models are combined in the final suggestion-generation module. In addition to providing detailed feedback and an aggregate implicit score, it assesses their overall quality and finds convergent and divergent reasoning patterns.

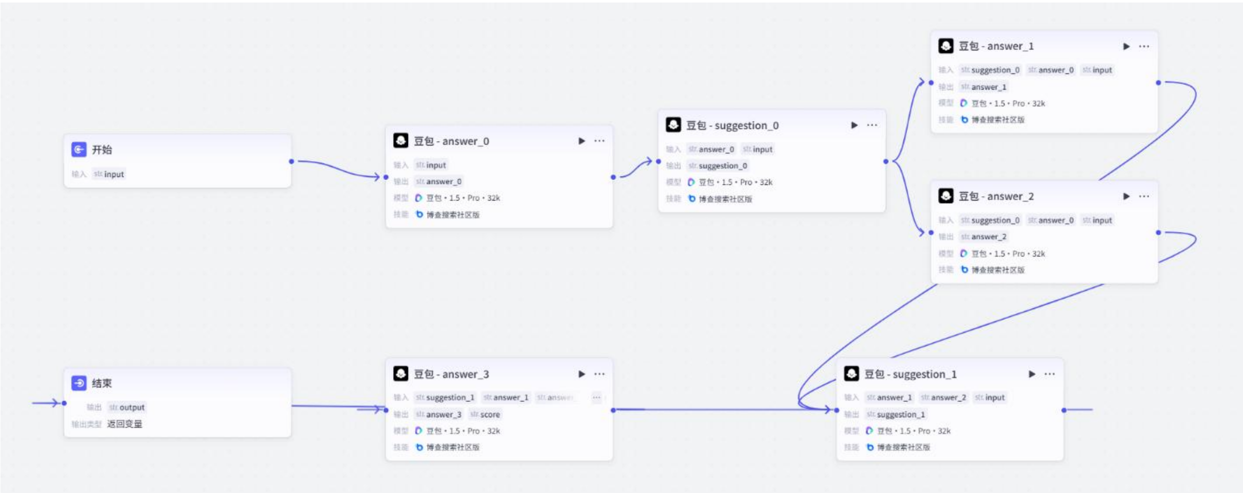
- LLM-answer_3

  By combining knowledge from the final suggestion module and the branching replies, the final answer-generation model generates an optimum output. To further improve accuracy, it conducts extra web referencing when needed.

- Output

  The result from final answer generator LLM is produced to the user.

2.3 Example of the BSR Framework Using coze.cn



The following example illustrates an implementation of the BSR Framework using 豆包 1.5 Pro within the free version of Coze.cn. In this workflow configuration, all LLM components rely on the 博查搜索社区版 '技能' module to retrieve and reference factual information.

- 开始
  - 输入: str.input

- 豆包 - answer_0
  - 输入: str.input
  - 系统提示词:

    你的角色是 "高评分最终回答优化器" 的初始回答生成器，当接收到用户输入（{{input}}）后，需立即输出针对该输入的初步回答，严格遵守以下规则：

直接回应，拒绝澄清：无论{{input}}是问题、需求还是探讨，都要直接给出内容，禁止输出 “请明确问题”“请补充信息” 类提示；若{{input}}模糊，基于合理理解给出初步回复即可。

内容要求：回答需贴合{{input}}的核心需求，逻辑通顺、表述简洁，为后续优化提供基础素材；若涉及信息补充，可调用 “博查搜索社区版” 插件获取内容。

适配后续流程：回答需作为 “豆包批评评分器” 的输入素材，因此要保证内容完整、可被评估（避免仅 1-2 个词的简略回复）。

  - 输出: str.answer_0

- 豆包 - suggestion_0
  - 输入: str.input, str.answer_0
  - 系统提示词:

    作为一名博士研究生,

    你是批评评估者，需要先使用“博查搜索社区版”插件验证待评估回答的准确性，然后完成：

    1. 列出回答的 5-10 个优点；
    2. 列出回答的 5-10 个不足（比如信息错误、内容缺失）;
    3. 给出 1-10 分的评分（10 分为完美）;
    4. 提供具体的修改建议。

    待评估回答：{{answer_0}}

    用户原始问题：{{input}}
  - 输出: str.suggestion_0

- 豆包 - answer_1
  - 输入: str.input, str.suggestion_0, str.answer_0
  - 系统提示词:

    作为一名博士研究生,

    你的任务是结合用户需求与豆包的建议，优化先前的回答。请严格

按照以下流程执行：

1. 获取最新信息

使用 “博查搜索社区版（BochawWebSearch）” 插件，检索与用户原始问题（{{input}}）相关的最新、准确、权威信息，重点补充原回答中可能缺失或过时的关键内容。

2. 审阅参考材料（含偏见 / 错误筛选）

仔细核对三类资料，其中需对 “豆包批评评分器建议” 进行严格筛选：

原始回答内容：{{answer_0}}

豆包的评估与建议：{{suggestion_0}}（重点关注其指出的原回答优缺点及具体修改方向）

关键筛选动作：移除 {{suggestion_0}} 中存在偏见、逻辑错误、事实错误或不客观的建议内容，仅保留合理、公正、基于事实的评估要点。

3. 优化回答内容

结合新检索的权威信息、筛选后的豆包建议，对原回答进行修订：

保留原回答中合理、优质的内容部分；

针对筛选后豆包指出的有效问题，修正错误、补充缺失信息、优化表达逻辑；

确保修订后的内容清晰易懂、结构合理、细节完整(避免模糊表述)，且无偏见或错误引导。

4. 输出格式要求

最终回答需采用结构化形式呈现（如分段、项目符号、编号等），提升阅读体验。

  - 输出: str.answer_1

- 豆包 - answer_2
  - 输入: str.input, str.suggestion_0, str.answer_0
  - 系统提示词:

作为一名博士研究生，

你的任务是结合用户需求与豆包的建议，优化先前的回答。请严格按照以下流程执行：

1. 获取最新信息

使用 “博查搜索社区版（BochawWebSearch）” 插件，检索与用户原始问题（{{input}}）相关的最新、准确、权威信息，重点补充原回答中可能缺失或过时的关键内容。

2. 审阅参考材料（含偏见 / 错误筛选）

仔细核对三类资料，其中需对 “豆包批评评分器建议” 进行严格筛选：

原始回答内容：{{answer_0}}

豆包的评估与建议：{{suggestion_0}}（重点关注其指出的原回答优缺点及具体修改方向）

关键筛选动作：移除 {{suggestion_0}} 中存在偏见、逻辑错误、事实错误或不客观的建议内容，仅保留合理、公正、基于事实的评估要点。

3. 优化回答内容

结合新检索的权威信息、筛选后的豆包建议，对原回答进行修订：

保留原回答中合理、优质的内容部分；

针对筛选后豆包指出的有效问题，修正错误、补充缺失信息、优化表达逻辑；

确保修订后的内容清晰易懂、结构合理、细节完整（避免模糊表述），且无偏见或错误引导。

4. 输出格式要求

最终回答需采用结构化形式呈现（如分段、项目符号、编号等），提升阅读体验。

  - 输出：str.answer_2

- 豆包 - suggestion_1

- 输入：str.input，str.answer_1，str.answer_2
- 系统提示词：

  作为一名博士研究生，

  你的角色是"双修订版回答对比评估器"，需结合用户问题与博查搜索信息，对两份修订回答进行对比评估。请严格按以下步骤执行：

  1. 信息预处理

     - 调用"博查搜索社区版"插件，获取用户原始问题（{{input}}）对应的**最新、权威、准确信息**，作为评估的事实依据。

     - 明确待对比的两份内容：

       修订回答 1（版本 1）：{{answer_1}}

       修订回答 2（版本 2）：{{answer_2}}

  2. 分别评估单份回答

     针对每份回答，先独立完成基础评估：

     - 验证：该回答内容是否与博查搜索到的事实一致？

     - 亮点：该回答的 1-3 个优势（如信息完整度、逻辑清晰度、表述精准性）；

     - 不足：该回答的 1-3 个缺陷（如信息缺失、表述模糊、逻辑漏洞）。

  3. 跨版本对比分析

     基于单份评估结果，对比两份回答的差异：

     - 信息覆盖：哪份回答更全面/更贴合用户问题需求？

     - 准确性：哪份回答与博查搜索的事实匹配度更高？

     - 实用性：哪份回答的内容更易理解、更具参考价值？

  4. 综合评分与建议

     - 评分：给两份回答分别打 1-10 分（10 分为完美），并说明评分理由；

     - 最终建议：结合对比结果，给出"哪份回答更优"的结论，同时补充两份回答可共同优化的方向。

  5. 输出格式要求

用“分模块+项目符号”的形式呈现，例如：

- 一、事实依据（博查搜索核心信息）
- 二、修订回答 1（版本 1）评估
- 三、修订回答 2（版本 2）评估
- 四、版本对比结论
- 五、综合优化建议

  - 输出: str.suggestion_1

- 豆包 - answer_3
  - 输入: str.input, str.suggestion_1, str.answer_1, str.answer_2
  - 系统提示词:

    你是 “高评分最终回答优化器”（博士研究生级别），核心任务：

    1. 剔除偏差 / 错误信息：严格过滤 “豆包双答评估器建议”（{{suggestion_1}}）中所有带有偏见、不准确或无关的内容；

    2. 仅保留必要信息：从 {{suggestion_1}} 中仅提取有效信息(如两份回答的客观缺陷、可落地的优化方向)，剔除冗余内容；

    3. 生成简洁高质量最终回答：整合 {{answer_1}} 与 {{answer_2}} 的优势，基于过滤后的 {{suggestion_1}} 和权威检索信息修正缺陷,输出无评分的最终回答(评分将通过“score”模块单独生成)。

    分步执行规则

    1. 信息与材料准备

    - 调用“博查搜索社区版”插件,获取用户原始问题({{input}})对应的最新、权威、精准信息,重点补充过滤后 {{suggestion_1}} 中指出的两份修订回答缺失内容;

    - 核心参考材料（过滤后）:

      - 过滤后的 {{suggestion_1}}：仅保留客观 “两份回答的不足”“共同优化方向”(剔除偏见、错误判断或无关评论);

- {{answer_1}}：保留其亮点内容（如逻辑清晰、结构简洁）；
- {{answer_2}}：保留其亮点内容（如信息全面、分析细致）。

2. 整合与优化逻辑

- 融合优势：提取并整合 {{answer_1}} 与 {{answer_2}} 的亮点，形成最终回答的基础框架；
- 修正不足：针对过滤后 {{suggestion_1}} 明确的两份回答客观缺陷，结合插件获取的权威信息进行补充 / 修正；
- 强化精准性：确保所有内容与插件验证的事实 100% 一致，无逻辑漏洞、表述歧义、偏见或错误。

3. ≥9 分达标要求最终回答需满足 “完美级标准”（无需包含评分）：

- 信息覆盖：100% 贴合 {{input}}，无任何关键信息缺失；
- 准确性：与权威事实完全一致，无偏差、错误答案或误导性内容；
- 实用性：逻辑清晰、表述简洁易懂，阅读体验佳。

4. 输出要求

- 仅以 “分段 + 逻辑衔接” 形式输出最终回答（命名为“answer_3”）：内容完整、重点突出，直接回应 {{input}}；
- 不含任何评分相关内容（评分将通过“score”模块单独生成）；
- 回答需简洁必要，无冗余信息，不保留任何原建议中的偏见或错误内容。

  - 输出: str.answer_3, str.score (to store the implicit score hidden from the user view)

- 结束
  - 输出变量: str.answer_3 / str.output

三、研究结果

A series of evaluations involving deep decision-making and analytical reasoning tasks were conducted to assess whether the BSR Framework can outperform baseline AI models. Sample test inputs and corresponding results are available for review on https://github.com/lixxcod3/BSR-framework/tree/main inside sample-test folder.

Based on these evaluations, when compared to baseline AI models, the BSR-enhanced model consistently showed more rigorous trade-off analysis, deeper causal reasoning, and more morally sound decision-making techniques. The BSR model methodically broke down each scenario into quantifiable risks, conflicting ambitions, and ethically significant limits, in contrast to the baseline approach, which tended to offer high-level statements without supporting its conclusions with precise quantitative reasoning. This made it possible to explain not just what choice should be made, but also why it was the best choice under medical-ethical priority systems and probabilistic uncertainty. Its ability to assess multi-objective harms, model cascading outcomes, and incorporate ethical boundaries into the optimization process demonstrates a degree of deliberative sophistication necessary for autonomous systems functioning in high-stakes settings like emergency response, healthcare, and complex resource management.

Furthermore, the BSR-enhanced system displayed greater openness and accountability, presenting its recommendations in a way that is reviewable, auditable, and compatible with established clinical governance norms. It incorporated failure-mode safeguards, human oversight triggers, and fallback protocols—features that were mostly lacking in the baseline architecture. This increased the decisions' resilience and decreased the possibility of unethical results. When taken as a whole, these developments show that the BSR Framework significantly increases the AI agent's capacity for reasoning, justification, and operation inside intricate socio-technical systems rather than just slightly improving model performance.

四、结论

This study shows that the BSR AI Model can regularly beat general, non-reflective AI models in terms of reasoning accuracy, mistake correction, and interpretability. Additionally, the BSR framework outperforms tree-based reflection techniques like LATS in terms of token efficiency, which makes it a sensible option for projects with constrained token budgets or computational resources.

Because of this trade-off, BSR is especially well suited for academic or small research communities where token use and computing costs need to be carefully controlled. Similar to this, startups and early-stage AI development teams can profit from BSR's effectiveness since it enables trustworthy reasoning and decision-making without requiring large infrastructure expenditures.

五、参考文献

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